

Presence and Passage

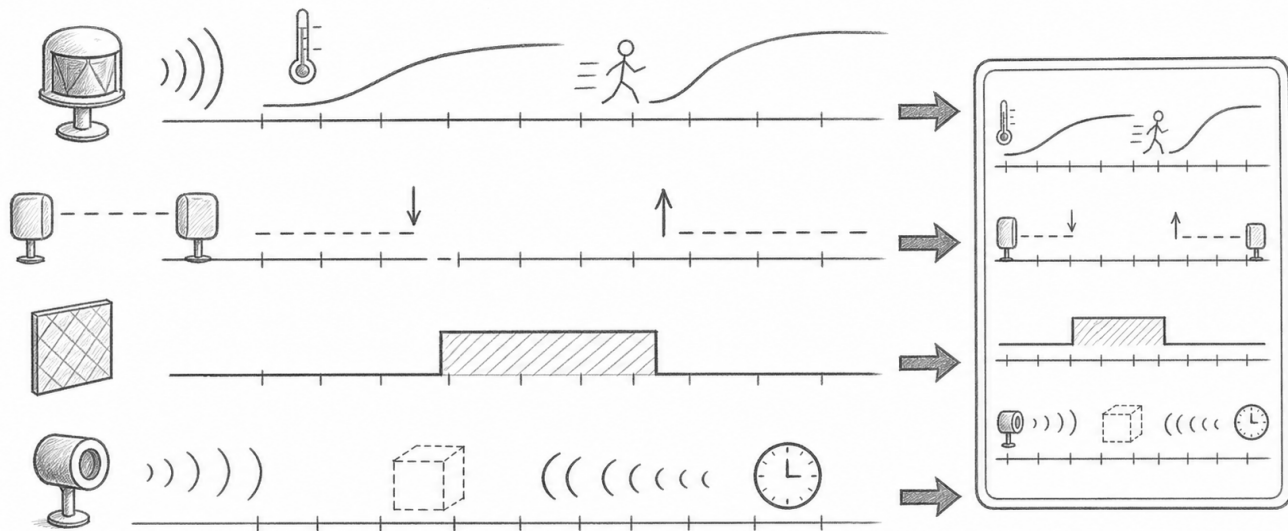
Preserve source identity, age, and disagreement

Host-verified E0 replay; powered composition and bench gates remain open

Lesson	041 — component	Time	120–180 minutes
Prerequisites	Lessons 019 and 040	Board	synthetic Mega replay only
Evidence	D30 acquisition LED and D31 TP-PRESENCE LED	Status	host verified E0; powered composition and bench open

Scope gate. This lesson composes supplied observations. It publishes no supported sensor circuit, wiring table, formal schematic, or powered adapter. Family names authorize planning only. It makes no lux, distance-accuracy, occupancy, security, navigation, collision-avoidance, traffic-enforcement, identity, or life-safety claim.

1 Keep four evidence lanes separate



Alternative text: four hand-drawn timelines retain separate PIR, beam, reflective-marker, and ultrasonic histories before entering one copied evidence frame. Their different event epochs show that the values are not assumed simultaneous.

`PirObservationPolicy` qualifies copied digital samples. Warm-up must finish before motion becomes eligible. Continuous qualified motion can become `StuckMotion`; that is retained evidence, not proof that a person is present. Clear and recovery use independent dwell intervals.

`PresenceModel` accepts already-qualified PIR and optical values plus a copied `TimedRangeEvidence`. It does not resample, requalify, vote, or replace a source. The E0 policies own no endpoint, claim, bus, timer, interrupt, emitter, heap, callback, or hidden clock.

Retained evidence	Meaning
PIR phase, epoch, age	warm-up, clear, motion, stuck motion, or source fault
beam provenance and edges	qualified interruption and restoration evidence
finish-guard provenance	calibration-bound reflective evidence from Lesson 040
range acquisition and copy epochs	measurement interval remains distinct from course time
quality, status, and age	old evidence remains stale rather than becoming false

2 Age is part of the value

Every present source epoch must be at or before the supplied frame time under unsigned modular subtraction, and its age must stay below half-range. A future epoch, exact half-range, or larger apparent age is a timing fault before ordinary mutation.

A required absent source yields **Unqualified** with an Ok status; absence is not invented hardware failure. Present malformed evidence faults. Freshness is evaluated per source, so one newer observation cannot refresh an older one.

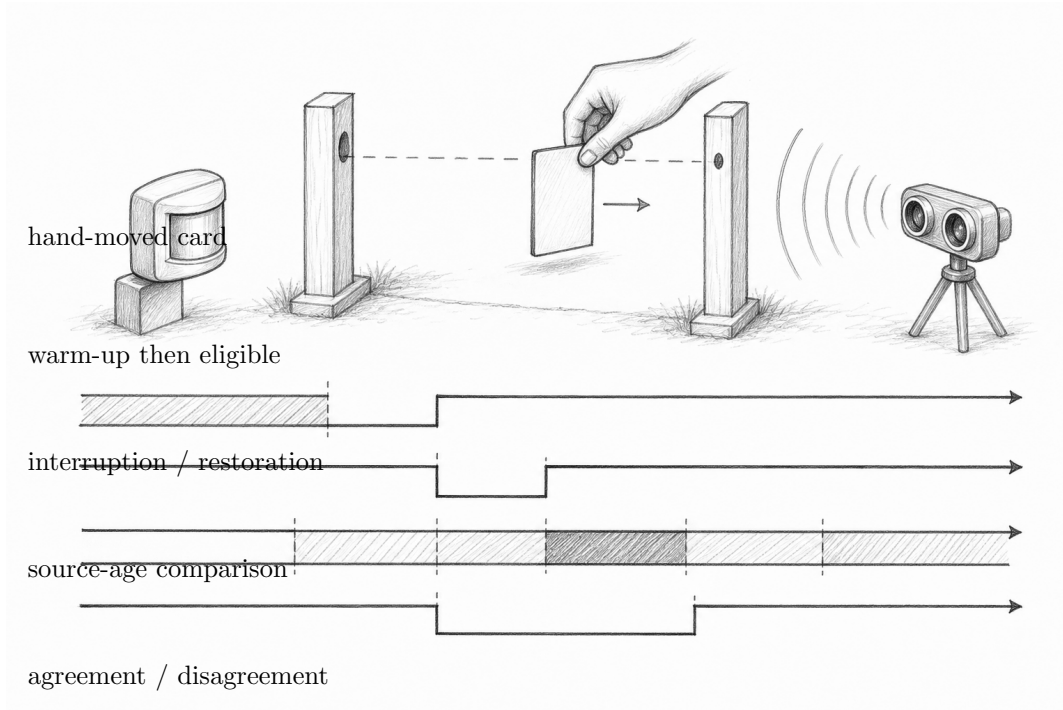
Range evidence deliberately names two domains. The microsecond acquisition interval is the pair

`measurementStartedAt` and `measurementLatency`.

The course-clock fields `startedAt` and `completedAt` identify when acquisition began and when the terminal outcome was copied. A completion stamp does not move the measurement epoch or pretend it was simultaneous with newer optical evidence.

3 Agreement is not a vote

Each configured agreement source contributes one boolean claim only after it is present, fresh, and valid. If the selected claims differ continuously through `agreementWindow`, the snapshot becomes **Disagreement**. Matching evidence clears that candidate. Missing, stale, or faulted required evidence keeps its own higher-priority classification instead of being mislabeled disagreement.



Alternative text: a hand moves an inert card between conceptual gate posts while separate motion, beam, age, and agreement timelines remain visible below. The drawing is orientation evidence, not wiring guidance.

4 One bounded passage event

A passage candidate begins only on a valid beam interruption. A valid restoration emits one `passageEvent` when it closes that candidate after positive time no greater than `beamPassageWindow`, while the complete required frame is fresh, valid, and not disagreeing. A long, stale, faulted, or disagreeing frame clears the candidate without emitting.

The beam owns the transition. PIR, finish-guard, and range values remain explicit context rather than a hidden majority vote. A repeated identical source event is deduplicated by its typed identity, provenance, event kind, and timestamp.

5 Four-frame predict, observe, interpret

1. Keep the canonical sketch in replay mode. Predict two clear frames, one active frame, then one restoration frame. The sketch supplies already-qualified `ReadyClear`, `Motion`, and `ReadyClear` PIR observations; it does not run PIR warm-up.
2. Observe D30 turn on only after both LED resources and the policy initialize. This is the acquisition signal. Observe D31 (TP-PRESENCE) remain off for the first two clear frames, turn on for the active frame, and remain on for the restoration frame's passage event.
3. In the active frame, predict PIR eligibility, beam interruption, finish-guard activation, and a 250 mm valid approach. D31 presents their combined state; it does not identify which source is active.
4. In the restoration frame, predict clear PIR, restored beam, inactive guard, and 800 mm range, plus exactly one `passageEvent`. D31 remains on as passage evidence.

D30 and D31 each drive one LED through 1 k Ω to ground in the documented E1 presentation. D31 is selected off before policy initialization and shuts down to high impedance on the fault path. Resource acquisition and electrical safe state remain separate observations. The volatile `evidenceCell`, `qualityCell`, and `rangeCell`

are debugger-readable supporting detail, not additional non-Serial indicators.

6 Troubleshooting tree

1. **Unqualified:** identify the required absent source. Do not call absence a hardware fault.
2. **Stale:** compare each source epoch with the frame epoch. Do not refresh old evidence by copying it again.
3. **Disagreement:** inspect all selected claims and their ages. Do not choose a winner.
4. **Source fault:** preserve the original status and source identity.
5. **Timing fault:** check future, half-range, backward, and changed same-time evidence before interpreting physical meaning.
6. **No passage event:** check interruption history, positive duration, passage window, freshness, and disagreement in that order.

7 Open powered and acceptance record

Required evidence	Record
Exact markings, revision, and authoritative source for each specimen	_____
Pin order, supply, rails, polarity, pull-ups, emitter, and current	_____
Aggregate pin, current, cadence, SRAM, flash, and stack result	_____
Resource acquisition prediction / observation / interpretation	_____
Named test-point safe state after shutdown	_____
Board revision, instruments, tool version, date, and commit	_____

Leave every row blank until measured. Compilation, synthetic replay, a pencil drawing, and this document do not establish specimen conformance, electrical safety, powered behavior, or bench acceptance. Resource acquisition and safe electrical state require separate observations.

8 Software evidence and next use

Host tests exercise PIR warm-up, retrigger, stuck-motion recovery, every freshness boundary, disagreement, passage, completed-range tuple validation, same-time identity, rollover, half-range rejection, reset, and field-stable replay. Those are host-test exercises, not behavior demonstrated by the four-frame sketch. The E0 object gate measures `PresenceModel` at exactly 128 AVR bytes. The canonical Mega replay measures 8,818 bytes flash and 226 bytes static SRAM with the pinned AVR toolchain. Powered aggregate SRAM, stack, pin, current, and cadence evidence remains open.

Lesson 042 may consume the snapshot as course evidence. Its explicit debounced button press remains the sole start authorization; PIR motion can establish eligibility but can never start a run.

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